

Trainings ▾

Preparatory Steps

Single Turbocharger

- Remove the air crossover. Refer to Procedure 010-019 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-019-tr.html).
- Disconnect the oil supply hose. Refer to Procedure 010-046 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-046-tr.html).
- Disconnect the oil drain hose. Refer to Procedure 010-045 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html).



Multiple Turbochargers

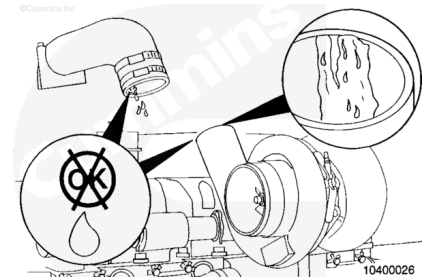
- Remove the air crossover. Refer to Procedure 010-019 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-019-tr.html).
- Remove the air transfer tube. Refer to Procedure 010-018 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-018-tr.html).
- Disconnect the oil supply hose. Refer to Procedure 010-046 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-046-tr.html).
- Disconnect the oil drain hose. Refer to Procedure 010-045 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html).



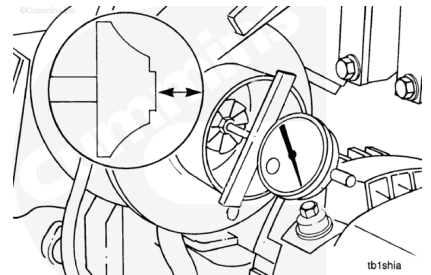
Initial Check

All Applications

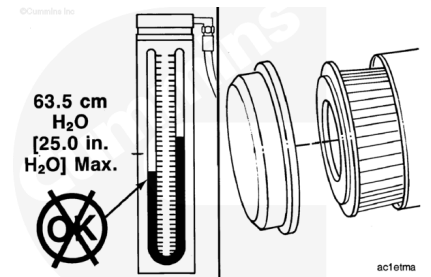
Inspect the compressor discharge and air crossover piping for oil.



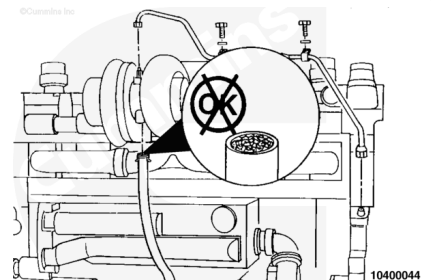
If oil is found, check the axial motion and radial clearance of the turbocharger, as outlined in the Measure section of this procedure.



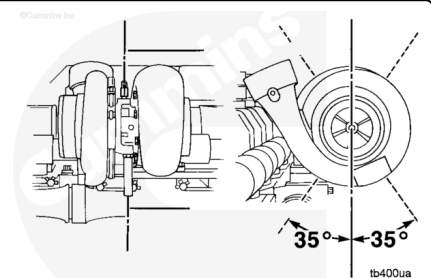
Check for intake restriction. Refer to Procedure 010-031 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-031.html).



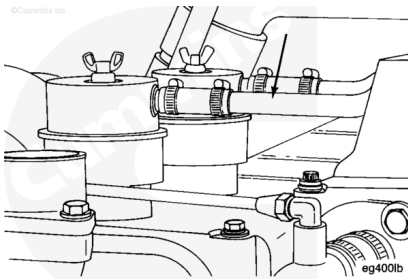
Check the oil drain tube for restrictions.
Clear any restrictions that are found.



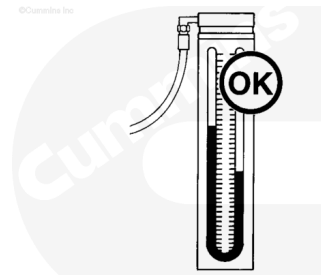
Check the angle of the drain tube.
The angle of the tube **must** be within 35 degrees of vertical.
Adjust the turbocharger, if necessary.



If the drain tube is free of restrictions and is positioned at the correct angle, check the crankcase breathers and tubes to make sure they are **not** plugged.

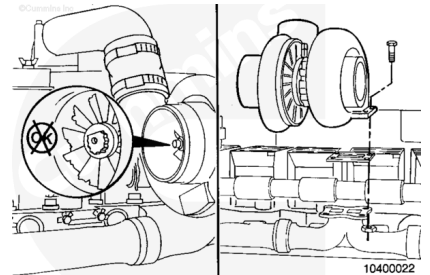


If these checks do **not** reveal the problem, measure the crankcase pressure (blowby). Refer to Procedure 014-005 in Section 14 (</qs3/pubsys2/xml/en/procedures/18/18-014-005.html>).



Inspect the turbocharger compressor impeller blades for damage.

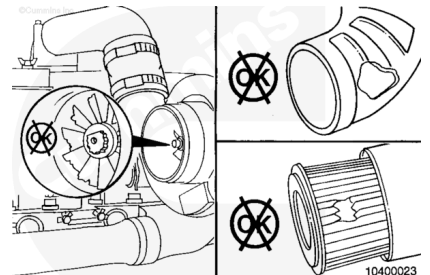
If the compressor impeller blades are damaged, the turbocharger **must** be replaced.



If the impeller is damaged, inspect the intake air piping and the filter element for damage.

Inspect the compressor cover for any debris.

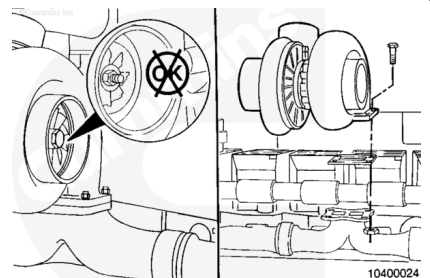
Repair any damage before operating the engine.



Inspect the turbine wheel for damage.

The turbocharger **must** be replaced if the turbine blades are damaged.

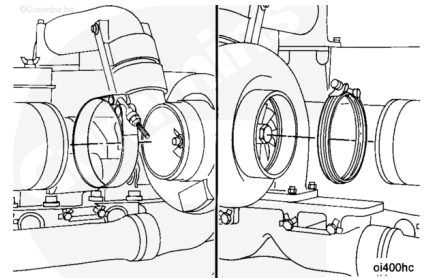
Check for debris stuck in the turbine housing.



Remove

Single Turbocharger

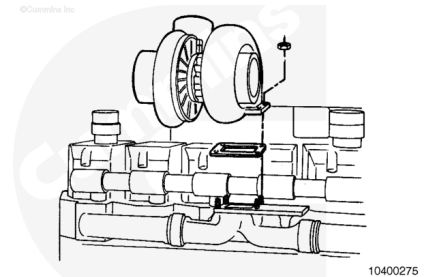
Remove the intake and exhaust pipes from the turbocharger.



Remove the turbocharger mounting nuts and washers.

Remove the turbocharger and gasket.

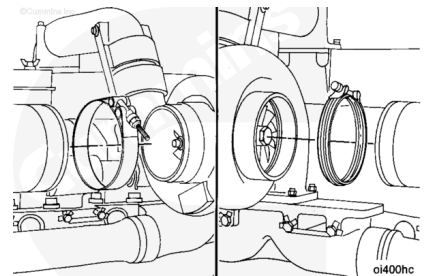
Discard the gasket.



Multiple Turbochargers

Remove the intake tubing from the low stage turbocharger.

Remove the exhaust connection from the low stage turbocharger.

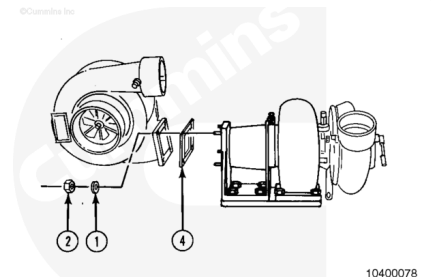


Tag the turbocharger mounting nuts and washers for future identification. The nuts are designed for high strength.

Remove the turbocharger mounting nuts (2) and washers (1).

Remove the low stage turbocharger and gasket (4).

Discard the gasket.



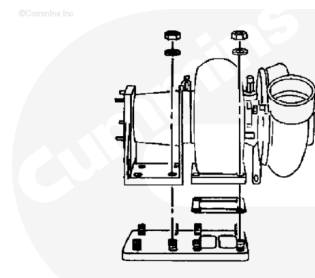
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

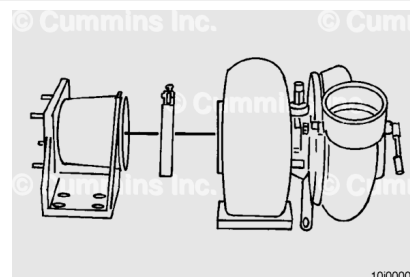
Remove the mounting nuts and washers.

Remove the high stage turbocharger and support assembly.

Remove and discard the gasket.



Loosen the v-band clamp and separate the high stage turbocharger and mounting support.



Inspect

Note : If inspections or dimensional checks indicate a problem, contact a Cummins® Authorized Repair Location. Reference the model number on the turbocharger dataplate.

If the engine is equipped with a turbocharger, inspect the turbocharger every 6000 hours or 2 years.

Remove the air intake and exhaust piping.

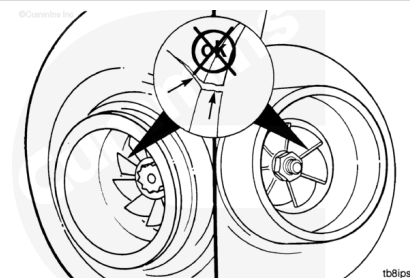
Check the turbocharger for damaged or cracked compressor or turbine blades.

Check to make sure the turbocharger shaft spins freely.

Check for any oil leaks on the turbine or compressor seals.

Refer to Procedure 010-040 in Section 10

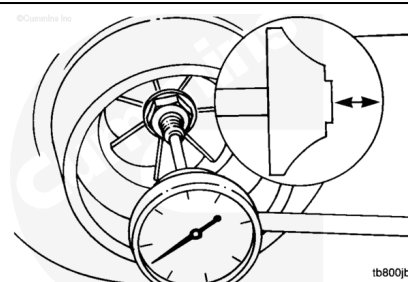
(</qs3/pubsys2/xml/en/procedures/18/18-010-040.html>) for compressor seal leaks. Refer to Procedure 010-049 in Section 10 (</qs3/pubsys2/xml/en/procedures/18/18-010-049.html>) for turbine seal leaks.



Measure

All Applications

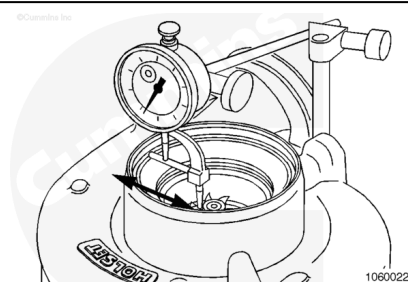
Measure the axial clearance (end-to-end) with a depth gauge, or a dial indicator, Part Number ST-537.



Axial Clearance			
	mm		in
HC5A/HX80	0.025	MIN	0.001
	0.152	MAX	0.006
T18A	0.10	MIN	0.004
	0.23	MAX	0.009

If the turbocharger is **not** within specifications, it **must** be replaced.

Measure the radial end play (side-to-side) at the compressor nose using a dial gauge.



Compressor Impeller Radial End Play			
	mm		in
Holset® HC5, HC5A (127 mm wheel)	0.229	MIN	0.009
	0.686	MAX	0.027
Holset® HC5, HC5A (130 mm wheel)	0.229	MIN	0.009
	0.762	MAX	0.03
HX80	0.254	MIN	0.01
	0.787	MAX	0.031

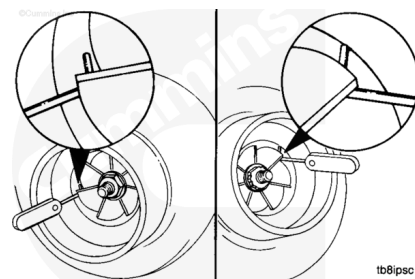
Make sure movement is within the MIN/MAX total indicator reading (TIR) values shown above.

Note : If the clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Contact a Cummins® Authorized Repair Location. See HC5 Turbocharger Shop Manual, Bulletin 3810243, for rebuild instructions.

Measure the radial clearance (side-to-side) with a wire-type feeler gauge.

Compressor Impeller and Turbine Wheel Radial Clearance

	mm		in
AiResearch [™] T18A	0.08	MIN	0.003
	0.18	MAX	0.007



Note : If the clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Contact a Cummins® Authorized Repair Location. See T35/50 VT50/ST50 Turbocharger, Bulletin 3379091, for rebuild instructions.

Install

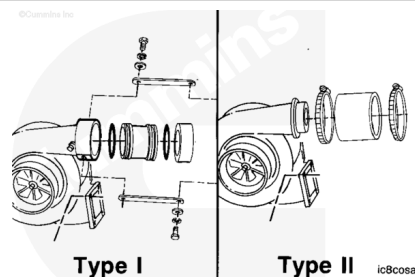
All Applications

Turbochargers are available with two types of compressor housings.

Type 1 - requires two o-rings, dust seal, connector pipe, and two retainer straps.

Type 2 - requires two hose clamps and a hose.

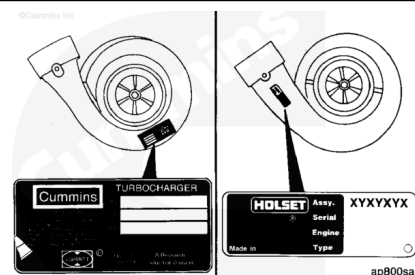
Changing from one type to another requires a new air crossover.



Two different turbochargers are used on the K19 engine: AiResearch[™] and Holset®.

Check the data tag on the turbocharger to determine the model.

The name is also cast on the housing of each turbocharger.

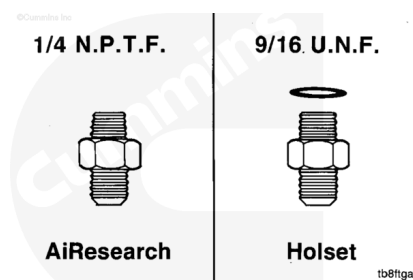


Two different types of oil supply fittings are used.

The AiResearch[™] turbocharger uses a ¼ NPTF (pipe plug) fitting.

The Holset® turbocharger uses a 9/16-18 UNF (straight thread) o-ring type fitting.

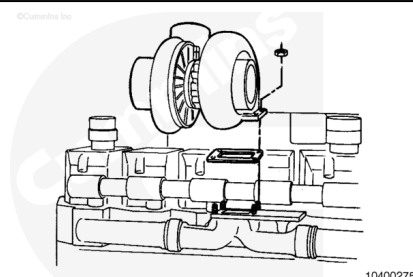
A new fitting is required if the model of turbocharger being installed is different from the original.



Single Turbocharger

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Apply anti-seize compound to the turbocharger mounting studs.

Install the gasket with the word OUT toward the turbocharger.

Install the turbocharger.

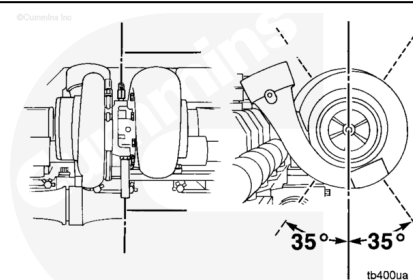
Install the washer and nuts.

Tighten the nuts.

Torque Value: 45 n•m [33 ft-lb]

Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

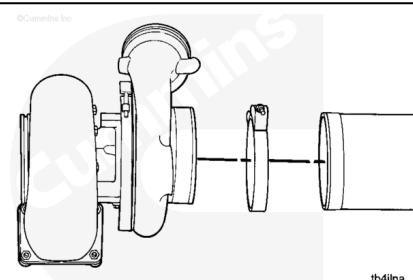
Some turbochargers use the through bolt style to attach the bearing housing to the turbine housing. Others use the clamp plate style.



Install the intake piping to the turbocharger.

Tighten the v-band clamp.

Torque Value: 9 n•m [80 in-lb]



Connect the exhaust piping to the turbocharger.

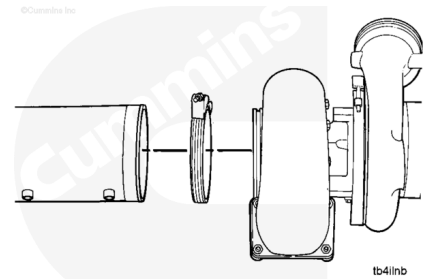
Tighten the v-band clamp.

Torque Value: 4 n•m [35 in-lb]

Some engines contain a flanged exhaust connection.

Tighten the capscrews.

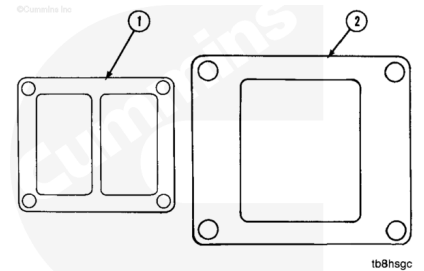
Torque Value: 45 n•m [33 ft-lb]



Multiple Turbochargers

⚠ CAUTION ⚠

The turbochargers and components must be installed in the sequence given. Failure to do so can result in component failure.

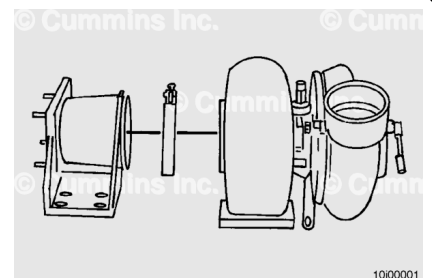


The high stage turbocharger (1) attaches directly to the exhaust manifold. This turbocharger has a divided turbine housing.

The mounting flange on the high stage turbocharger housing is much smaller than the low stage turbocharger.

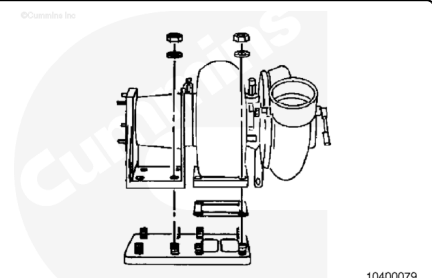
Install the mounting support and v-band clamp on the high stage turbocharger.

Tighten the clamp enough to hold the parts together and allow them to rotate.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Do **not** use anti-seize compound on the mounting studs or nuts.

Install the gasket on the exhaust manifold with the word OUT toward the turbocharger.

Install the turbocharger and support on the manifold.

Install the washers and nuts.

Hand-tighten the nuts **only**.

Check to make sure the turbocharger and the support fit flatly on the manifold.

Remove the v-band clamp.

Check the fit between the turbocharger and the support. Both parts **must** be in contact with the manifold and without gaps.

Install the v-band clamp and tighten.

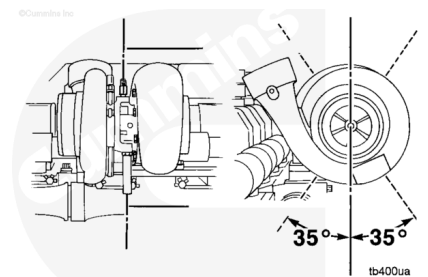
Torque Value:

AiResearch™ 7 n•m [62 in-lb]

Torque Value:

Holset® 14 n•m [124 in-lb]

Check the oil drain tube angle. The turbocharger drain tube **must** be within 35 degrees of vertical.



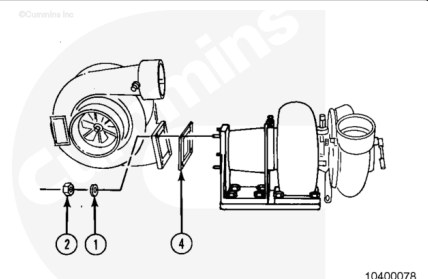
Do **not** use anti-seize compound on the studs or nuts.

Install the gasket (4) on the support assembly with the word OUT toward the low stage turbocharger.

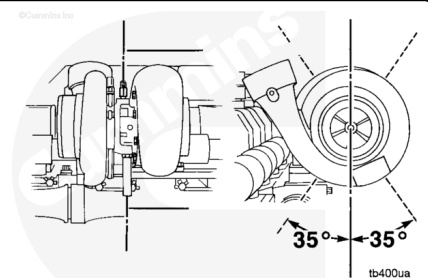
Install the low stage turbocharger, washers, and nuts.

Tighten the nuts.

Torque Value: 45 n•m [33 ft-lb]



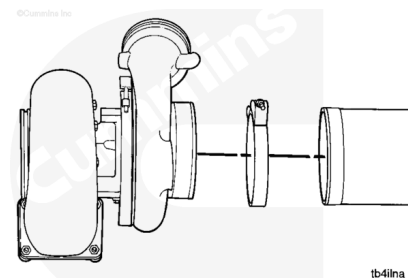
Check the oil drain tube angle. The turbocharger drain tube **must** be within 35 degrees of vertical.



Install the intake piping to the turbocharger.

Tighten the v-band clamp.

Torque Value: 9 n•m [80 in-lb]



Connect the exhaust piping to the turbocharger.

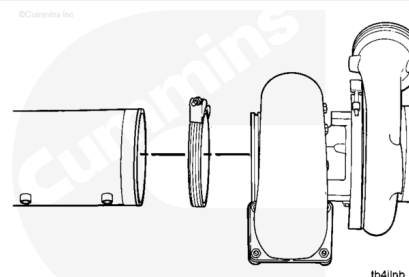
Tighten the v-band clamp.

Torque Value: 4 n•m [35 in-lb]

Some engines contain a flanged exhaust connection.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



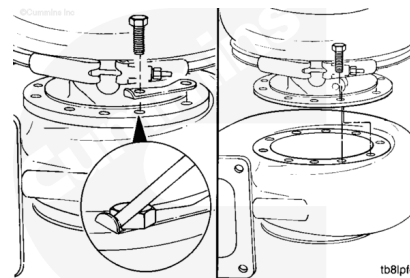
Adjust

All Applications

To adjust the through bolt style of turbocharger, bend the lockplates off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing to align the drain tube and capscrew holes.

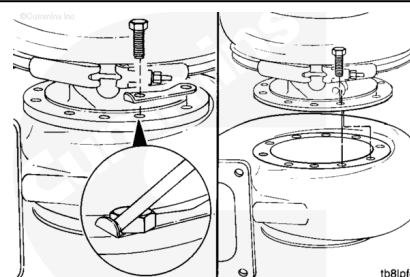


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 12 n•m [106 in-lb]

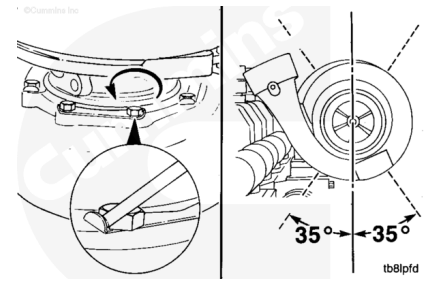
Bend the lockplate tabs over the capscrew heads.



To adjust the clamping plate style turbocharger, bend the lockplates off the capscrew heads.

Loosen all capscrews.

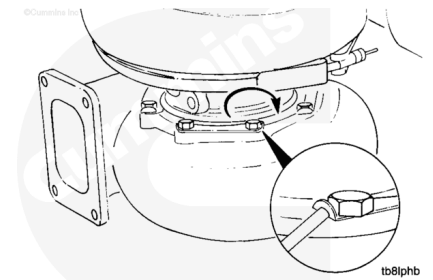
Rotate the bearing housing until the angle of the drain tube is correct.



Tighten the capscrews.

Torque Value: 12 n•m [106 in-lb]

Bend the lockplate tabs over the capscrew heads.



Finishing Steps

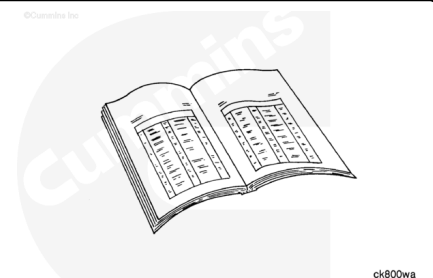
Single Turbocharger

- Connect the oil drain hose and fill the turbocharger bearing housing with engine oil. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html>).
- Connect the oil supply hose. Refer to Procedure 010-046 in Section 10 (</qs3/pubsys2/xml/en/procedures/18/18-010-046-tr.html>).
- Install the air crossover. Refer to Procedure 010-019 in Section 10 (</qs3/pubsys2/xml/en/procedures/18/18-010-019-tr.html>).



Multiple Turbochargers

- Connect the oil drain hose and fill the turbocharger bearing housing with engine oil. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html>).
- Install the air transfer tube. Refer to Procedure 010-018 in Section 10 (</qs3/pubsys2/xml/en/procedures/18/18-010-018-tr.html>).



- Connect the oil supply hose. Refer to Procedure 010-046 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-046-tr.html).
- Install the air crossover. Refer to Procedure 010-019 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-019-tr.html).

Last Modified: 18-Aug-2026
